

RESEARCH SYNOPSIS

Applicant: Swarup G L

Proposed Theme: Advanced AI for Healthcare & Scientific Systems

Target Mentor: Dr. Sudipta Roy

1. STATEMENT OF INTENT

My opinion on AI was, The true test of an artificial intelligence system lies not in its theoretical accuracy on a pristine dataset, but in its ability to function reliably under strict physical and computational constraints. As an undergraduate in Artificial Intelligence and Data Science at IIIT Raichur, my core engineering focus is on bridging complex algorithmic theory with deployable edge hardware. I am applying for the Summer Research Internship Programme 2026 at the Jio Institute to work under the mentorship of Dr. Sudipta Roy. My objective is to contribute to the "Advanced AI for Healthcare & Scientific Systems" theme by developing mathematically robust, highly compressed deep learning architectures capable of processing noisy clinical data directly on edge diagnostic devices.

2. RATIONALE FOR CHOOSING THIS THEME

Clinical imaging—such as ultrasound, MRI, and time-lapse biological monitoring—presents computational challenges that standard vision algorithms fail to address. Medical data is inherently plagued by low contrast, speckle noise, and severe class imbalances. Dr. Roy's pioneering research in developing multimodal AI frameworks and quantitative imaging biomarkers for maternal-fetal care represents the exact frontier I wish to explore.

I am drawn to this research because I believe the future of healthcare AI relies on edge-deployable solutions rather than cloud-dependent systems. Building robust diagnostic utilities for low-resource settings requires a synthesis of rigorous mathematical preprocessing and lightweight neural networks. Dr. Roy's laboratory provides the ideal environment to systematically tackle these constraints and translate raw biomedical signals into actionable clinical insights.

3. TECHNICAL PREPARATION AND ENGINEERING PHILOSOPHY

My approach to complex technical challenges is rooted in a "solve-to-learn" methodology: extracting the core principles of a problem and iteratively building functional systems to master the underlying mechanics. This philosophy has guided my research across edge computing, medical imaging, and multi-modal sensor fusion.

- **Embryo Morphokinetics:** Standard cross-entropy loss often fails to account for the chronological nature of biological development. To address this, I developed a CNN+LSTM sequence model for a 16-class embryo dataset. By engineering a custom Hybrid Ordinal Regression Loss that combined categorical accuracy with an expected-index MSE penalty, I explicitly taught the network chronological awareness, achieving an 89.42% tolerance accuracy (± 1 phase).
- **Edge-AI for Clinical Ultrasound (BUSI):** To combat the severe computational overhead of standard models, I engineered a 48K-parameter custom Depthwise Separable CNN from scratch to classify breast ultrasound tumors. Utilizing Global Average Pooling and progressive channel expansion, this architecture achieved an 83.33% peak accuracy while reducing the parameter footprint by 99% relative to VGG16 baselines, proving its viability for low-power medical tablets.

4. EXPECTED CONTRIBUTIONS

During the 8-week residency, I thought contribute :

1. Designing adaptive, multimodal preprocessing pipelines to filter scanner noise from raw clinical inputs before they reach the neural network.
2. Training and benchmarking highly compressed CNN and LSTM architectures for rapid diagnostic segmentation and temporal sequence analysis.